Mahmoud Alyosify

Academic CV · PhD applicant in Artificial Intelligence

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RESEARCH INTERESTS

Research interests: **efficient inference for large language models**, careful evaluation of generative AI, self-supervised representation learning, reinforcement learning for resource decisions, and multi-agent systems.

EDUCATION

MSc in Artificial Intelligence — Sep 2025 – Oct 2026
Queen's University — Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) — Sep 2019 – Jul 2023
Assiut University — Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

RESEARCH EXPERIENCE

MinimalLM — Master's thesis — 2026

Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- Question: can an unchanged LLM write fewer tokens and still give a complete answer? Tested on 330 questions (MT-Bench, AlpacaEval, Arena-Hard-Auto) and six other languages.
- Found that the simplest stop rule saves 32.6% but cuts off 59.3% of answers; added a guardrail instruction and a completeness check so the saving no longer costs the answer.
- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Compared six versions on 5,940 answers to show what each part adds: without the completeness check 17.3% of answers are cut off; without the instruction the saving falls to 19.8%.
- Honest limits: shorter answers more often leave out part of a broad question (22.9% vs 12.6% by the judge), and the evaluation is internal, with no independent human study.
- Built to be reproduced: pinned models and GPU, greedy decoding, 56 automated tests, and a notebook that recomputes every number from the stored answers.

Human vs. AI-Agent Pull Requests — research study

Python, GitHub data mining, statistics

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

SimCLR-Vision-SSL — self-supervised image model — 2026

PyTorch, SimCLR, SupCon, ONNX, FAISS

- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

RL-Cloud-Autoscaler — deep RL for cloud scaling — 2026

Python, Gymnasium, PPO, DQN

- Designed the environment from scratch: state, actions and a reward that penalises both overspending and slow responses.
- Compared both agents with static allocation and threshold-based autoscaling, the policies real teams run.

HORUS Sentinel — multi-agent intelligence platform 2026
Python, LLM agents, knowledge graph, self-hosted LLM

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.

TEACHING EXPERIENCE

Part-Time Machine Learning Instructor Jul 2025 – Present
National Telecommunication Institute (NTI) Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.
- Write the final exams and assess the trainees' end-to-end capstone projects.

Online Instructor 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROFESSIONAL EXPERIENCE

AI Engineer Intern Oct 2026 – Present
Injaz Tech (Injaz Group) Maadi, Cairo

- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20 through measured design changes.

AI in Cybersecurity Intern Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

AWARDS & SCHOLARSHIPS

- Digilians Presidential Scholarship for graduate study at Queen's University (2025)
- Semi-Finalist, Microsoft Imagine Cup 2023 · Finalist, ISEIC 2023
- 1st Place, Smart Cities Hackathon 2022 · 1st Place, Science & Technology Content Competition
- Ideal Student Award, Faculty of Computers & Information, Assiut University (2022–2023)

CERTIFICATIONS

- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI)

SKILLS

Methods: experiment design, ablation studies, statistical testing (McNemar, bootstrap), reproducible benchmarks
Tools: PyTorch, Hugging Face Transformers, scikit-learn, Gymnasium, FAISS, ONNX
Computing: Python, C++, SQL, Git, Linux, NVIDIA A40, AWS

LANGUAGES

Arabic (native) · English (professional working proficiency) · References available on request